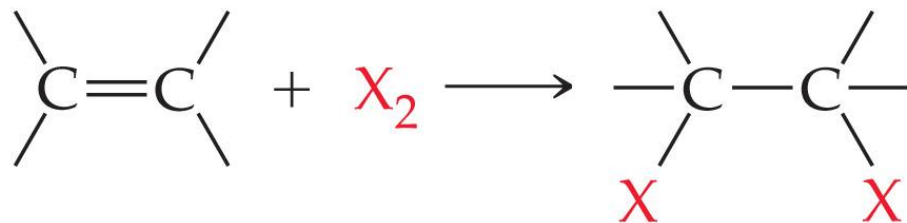


13.6 Addition Reactions of Alkenes

Addition of Cl_2 and Br_2 to Alkenes and Alkynes: Halogenation

- Alkenes and alkynes react with halogens to yield 1,2-dihaloalkane addition products:



(A 1,2-dihaloalkane where $\text{X} = \text{Br}$ or Cl)

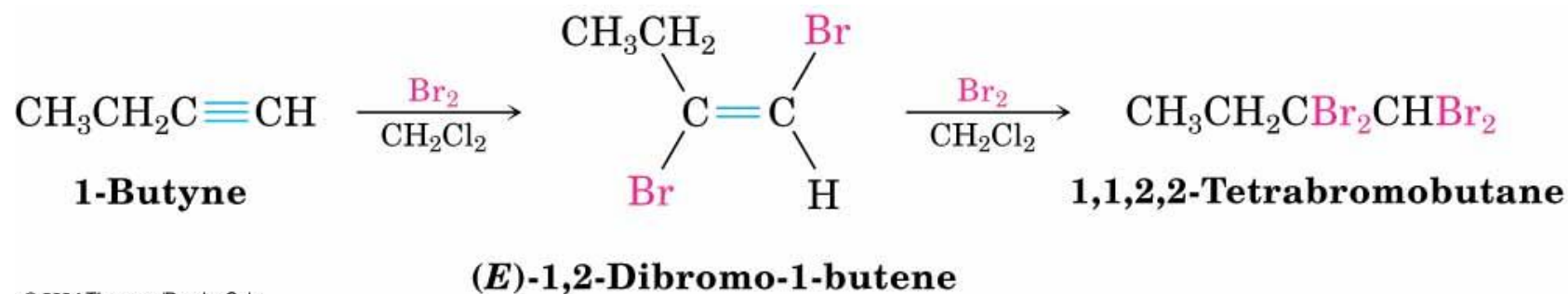
Bromine (Br_2) can be used as a test for multiple bonds and unsaturation of fats.



(a)

Addition of Bromine and Chlorine

- Initial addition gives trans intermediate
- Product with excess reagent is tetrahalide

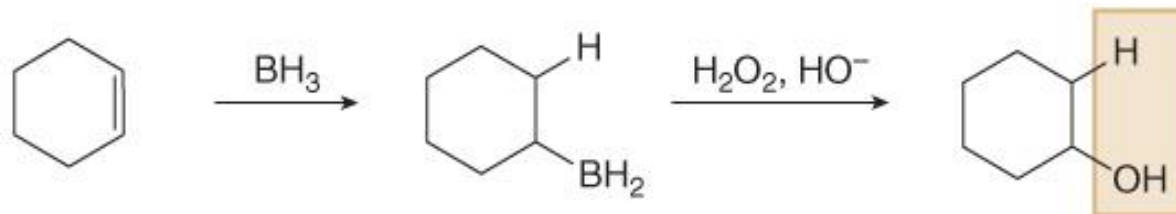
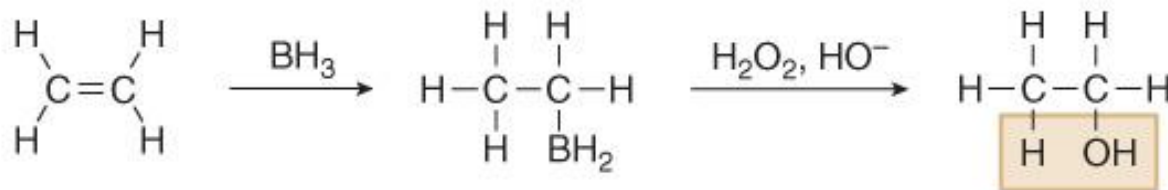


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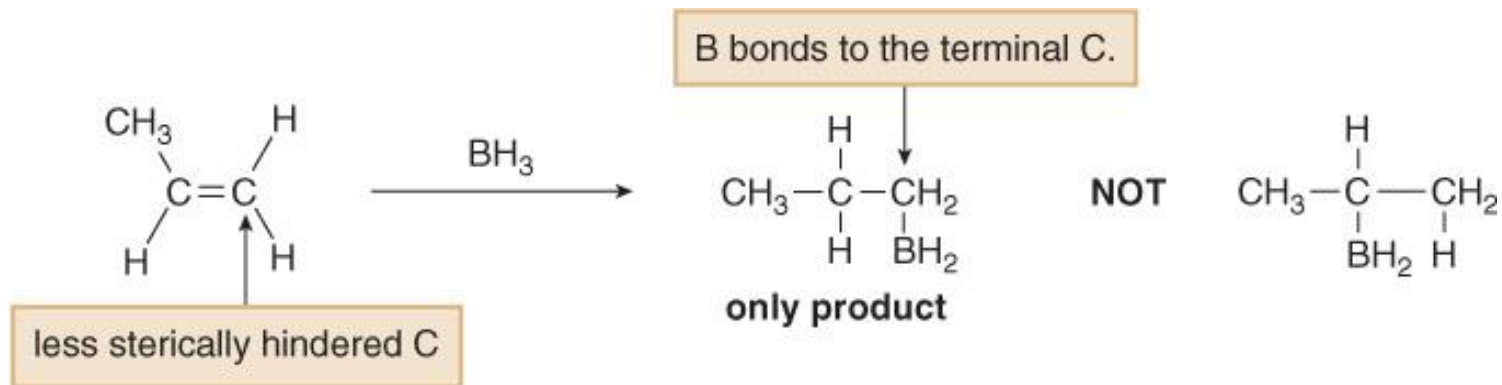
3.13- Hydroboration - Oxidation:

- Hydroboration—oxidation results in the indirect addition of H_2O to an alkene. **Anti Markovnikov Addition**

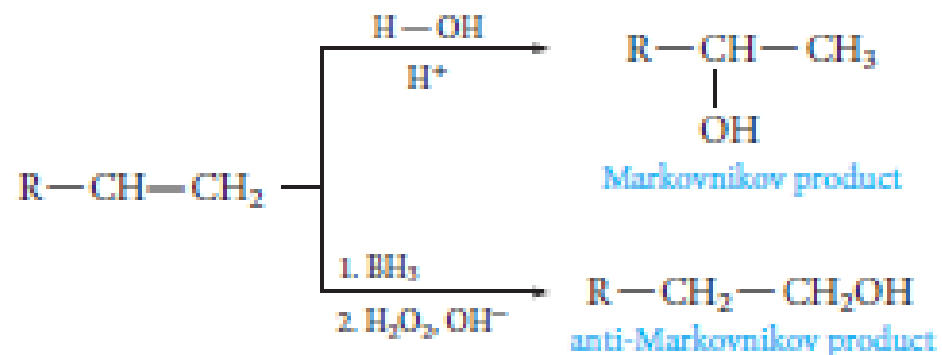
Examples



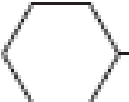
- With **unsymmetrical alkenes**, the boron atom bonds to the less substituted carbon atom.



One great advantage of this hydroboration–oxidation sequence is that it provides a route to alcohols that *cannot* be obtained by the acid-catalyzed hydration of alkenes



PROBLEM 3.19 What alcohol is obtained by applying the hydroboration–oxidation sequence to 2-methyl-2-hexene?

PROBLEM 3.20 What alkene is needed to obtain -CH₂CH₂OH

via the hydroboration–oxidation sequence? What product would this alkene give with acid-catalyzed hydration?